

KUSETTI TARUN

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Summary

Electronics and Communication Engineering student with hands-on knowledge of C programming, Embedded Systems, and STM32 microcontrollers. Skilled in interfacing and programming STM32 peripherals, sensor integration, and IoT-based project development. Capable of designing, debugging, and implementing real-time embedded applications with a strong interest in embedded firmware development, automation, and hardware-software integration.

Education

Bachelor of Technologies (ECE) , <i>Dr. Lankapalli Bullayya College of Engineering</i> CGPA: 8.01%	2022 – 2026 Visakhapatnam
Intermediate (MPC) , <i>Gnanajyothi Junior College</i> Percentage: 87.6%	2020 – 2022 Srikakulam
Gayatri High School Percentage: 95%	2020 Srikakulam

Skills

- **Languages:** C, Embedded C, Python
- **Microcontrollers:** STM32, Arduino, ESP32
- **Embedded Concepts:** GPIO, UART, I2C, SPI, Timers, ADC, PWM, Peripheral Interfacing
- **Tools & IDEs:** STM32CubeIDE, Arduino IDE, KiCad
- **Operating System:** Linux (Ubuntu)
- **Cloud Platform:** Rightech Cloud
- **Embedded Software:** Linux Internals, Embedded Linux(Rugged Board), Zephyr RTOS

Internships

Industrial Training – BHEL (Bharat Heavy Electricals Limited)

Duration: 1 Month

Gained basic understanding of CNC machinery and its applications in industrial manufacturing. Observed real-time automation processes and learned about production workflows and safety standards.

Certificates

- Industrial Training at Bharat Heavy Electricals Limited.
- NPTEL - Micro Sensors and Nano Sensors

Projects

Autonomous Mobile Robot Based IOT Integration

Technologies: STM32CubeIDE, Embedded C, UART, PWM, Wi-Fi, IoT.

- Developed an IoT-based autonomous surveillance robot using **STM32F446RE** for real-time monitoring and control.
- Integrated **ultrasonic and flame sensors** using GPIO for obstacle detection and hazard identification.
- Configured and interfaced **ESP32-CAM** module using UART for wireless video streaming.
- Implemented **Wi-Fi-based communication** for remote monitoring and control.
- Controlled robot movement using **L298N motor driver** and PWM signals.
- Enabled **servo motor-based camera positioning** and **buzzer alerts** for safety indication.

Traffic Light Control System

Technologies: STM32CubeIDE, Embedded C, GPIO, Timers, UART

- Designed and developed an automated traffic light control system using the **STM32F446RE** microcontroller.
- Implemented a **Finite State Machine (FSM)** to control Red, Yellow, and Green signal transitions with predefined timing.
- Configured **hardware timers** to generate accurate one-second intervals for traffic signal operation.
- Interfaced a **7-segment display** to provide a real-time countdown for each traffic signal state.
- Utilized **GPIO** to control LEDs representing traffic signals and drive the display interface.
- Developed modular firmware using the **STM32 HAL** library, improving code readability, scalability, and maintainability.
- Verified signal timing accuracy and state transitions through testing and debugging.

Environmental Monitoring System using Zephyr RTOS

- Technologies: Zephyr RTOS, Embedded C, STM32 Nucleo-F446RE, BME280, I2C, UART, WE-10, RightTech IoT Cloud
- Developed an environmental monitoring application using Zephyr RTOS on the STM32 Nucleo-F446RE board by interfacing the BME280 sensor over I2C.
- Configured hardware using DeviceTree overlays and implemented sensor data acquisition using Zephyr device drivers.

Transmitted real-time temperature, humidity, and pressure data to RightTech IoT Cloud through the WE-10 Wi-Fi module using UART.

Distance Monitoring System

Technologies: STM32CubeIDE, Embedded C, UART, Timer Input Capture, MQTT, Right Tech IoT Cloud

- Developed an IoT-based distance monitoring system using STM32F446RE and the HC-SR04 ultrasonic sensor for real-time distance measurement.
- Implemented Timer Input Capture to accurately calculate distance and interfaced the WE10 Wi-Fi module via UART for wireless communication.
- Transmitted real-time distance data to Right Tech IoT Cloud using the MQTT protocol for remote monitoring and visualization.

PERSONAL DETAILS

Father's Name: K. Chinna Rao

Mother's Name: K. Parvathi

Date of Birth: 11/03/2004

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